

ShopFlow AWS Project

SQS Design

Message Queues - Dead-Letter Queues - FIFO for Payments

Module 12 of 18

Field	Value
Version	1.0
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Region	ap-south-1 (Mumbai)
Author	Manish Kumar Sharma
Application	ShopVerse -- Spring Boot 3.2.5 / Java 21

1. Overview

Amazon SQS provides durable, managed message queues for ShopVerse's asynchronous processing. SQS replaces RabbitMQ for the AWS deployment. A producer puts messages in the queue; a consumer processes them independently. If processing fails, messages are automatically retried. Failed messages move to a Dead-Letter Queue (DLQ).

2. Queue Inventory

Queue Name	Type	Purpose	Max Retries
shopflow-notifications-dev	Standard	Order confirmation emails, shipping updates	3
shopflow-notification-dlq-dev	Standard	DLQ for failed email messages	--
shopflow-payments-dev.fifo	FIFO	Stripe/PayPal webhook callbacks (exact-once)	3
shopflow-payment-dlq-dev.fifo	FIFO	DLQ for failed payment messages	--
shopflow-order-events-dev	Standard	Order placed, shipped, cancelled events	5
shopflow-order-events-dlq-dev	Standard	DLQ for failed order event messages	--

3. Standard vs FIFO Queue

Attribute	Standard	FIFO
Ordering	Best-effort	Strict first-in, first-out
Delivery	At-least-once	Exactly-once
Throughput	Unlimited	3,000 messages/second
Name suffix	No requirement	Must end with .fifo
ShopFlow use	Notifications, events	Payments (cannot double-charge)

4. Dead-Letter Queue (DLQ) Pattern

Queue Attribute	Main Queue	DLQ
Retention Period	4 days (345,600s)	14 days (1,209,600s)
Max Receive Count	3-5 attempts	N/A (final destination)
Action on failure	Redirect to DLQ	Alert ops team (CloudWatch alarm)

5. Message Processing Flow

Step	Action
1	Spring Boot puts order event JSON message into queue
2	Lambda polls the queue (up to 10 messages per batch)
3	Lambda processes each message (calls SendGrid / updates DB)
4	On success: Lambda deletes message from queue
5	On failure: message becomes visible again after VisibilityTimeout
6	After 3 failures: SQS moves message to DLQ
7	CloudWatch alarm fires when DLQ depth > 0

6. SQS vs RabbitMQ (Local ShopVerse)

Attribute	RabbitMQ (local)	SQS (AWS)
Management	Self-hosted Docker container	Fully managed, no server
Persistence	Disk-backed	Durable by default
Fan-out	Exchange + bindings	Via SNS to multiple SQS queues
Cost	EC2 instance cost	\$0.40/million messages